KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5123B | 5 Credits: 4 | Program Elective | Source-grounded

Quantity Surveying

 To provide knowledge on Quantity surveying and measuring principles To impart knowledge on Price Analysis

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5123B is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5123B.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Wood and Paper Technology
Course code	5123B
Course title	Quantity Surveying
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

11 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide knowledge on Quantity surveying and measuring principles To impart knowledge on Price Analysis
- To familiarize valuation of land and building
- To introduce computer applications in Quantity surveying

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding principles of measurements and description Comprehend analysis of price and understand	See official syllabus
CO2	14 Applying tendering and contracting	See official syllabus
CO3	Summarize the land evaluation and surveyor laws 15 Applying	See official syllabus
CO4	Describe Project management methods applications 14 Applying	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	description.	3	As specified
Module 2	Comprehend analysis of Price and understand tendering and contracting	3	As specified
Module 3	Summarize the land evaluation and surveyor laws	3	As specified
Module 4	Describe Project management methods and applications	2	As specified

MODULE 1

description.

This module is presented from the official syllabus scope for Quantity Surveying. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: description.
- Official duration: As specified
- Official topic entries captured: 3

- The full source excerpt is preserved below for coverage checking.

Explain basics of quantity surveying

Explain basics of quantity surveying is an official topic in Module 1 of Quantity Surveying. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain basics of quantity surveying using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain basics of quantity surveying should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain basics of quantity surveying means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Explain basics of quantity surveying [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain building estimates

Explain building estimates is an official topic in Module 1 of Quantity Surveying. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain building estimates using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain building estimates should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain building estimates means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Explain building estimates [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Develop an idea on measurement of work 5 Understanding Historical development of quantity surveying: Functions performed by the quantity surveyor in relation to construction works - Evolution of standard methods of measurement for construction Works. The use of document in practice. Basic concepts of quantity surveying: Measurement and computation of lengths, girths, areas, and volumes both for regular and irregular shapes from drawings. Theoretical processes of building contract from inception to completion and the interrelationship of the professional team Engineering drawings for buildings and various structures, Principles of standard methods of measurement of work. Bills of quantities, Drawings. Purpose of Bills of Quantities. Processes of preparing Bills of Quantities, including taking off, working up, abstracting and billing. Types of bill formats and their uses. Setting out of descriptive and quantitative information in taking off. Dimension, abstracting and Billing Sheets

Develop an idea on measurement of work 5 Understanding Historical development of quantity surveying: Functions performed by the quantity surveyor in relation to construction works - Evolution of standard methods of measurement for construction Works. The use of document in practice. Basic concepts of quantity surveying: Measurement and computation of lengths, girths, areas, and volumes both for regular and irregular shapes from drawings. Theoretical processes of building contract from inception to completion and the interrelationship of the professional team Engineering drawings for buildings and various structures, Principles of standard methods of measurement of work. Bills of quantities, Drawings. Purpose of Bills of Quantities. Processes of preparing Bills of Quantities, including taking off, working up, abstracting and billing. Types of bill formats and their uses. Setting out of descriptive and quantitative information in taking off. Dimension, abstracting and Billing Sheets is an official topic in Module 1 of Quantity Surveying. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop an idea on measurement of work 5 Understanding Historical development of quantity surveying: Functions performed by the quantity surveyor in relation to construction works - Evolution of standard methods of measurement for construction Works. The use of document in practice. Basic concepts of quantity surveying: Measurement and computation of lengths, girths, areas, and volumes both for regular and irregular shapes from drawings. Theoretical processes of building contract from inception to completion and the interrelationship of the professional team Engineering drawings for buildings and various structures, Principles of standard methods of measurement of work. Bills of quantities, Drawings. Purpose of Bills of Quantities. Processes of preparing Bills of Quantities, including taking off, working up, abstracting and billing. Types of bill formats and their uses. Setting out of descriptive and quantitative information in taking off. Dimension, abstracting and Billing Sheets using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop an idea on measurement of work 5 understanding historical development of quantity surveying: functions performed by the quantity surveyor in relation to construction works - evolution of standard methods of measurement for construction works. the use of document in practice. basic concepts of quantity surveying: measurement and computation of lengths, girths, areas, and volumes both for regular and irregular shapes from drawings. theoretical processes of building contract from inception to completion and the interrelationship of the professional team engineering drawings for buildings and various structures, principles of standard methods of measurement of work. bills of quantities, drawings. purpose of bills of quantities. processes of preparing bills of quantities, including taking off, working up, abstracting and billing. types of bill formats and their uses. setting out of descriptive and quantitative information in taking off. dimension, abstracting and billing sheets should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop an idea on measurement of work 5 Understanding Historical development of quantity surveying: Functions performed by the quantity surveyor in relation to construction works - Evolution of standard methods of measurement for construction Works. The use of document in practice. Basic concepts of quantity surveying: Measurement and computation of lengths, girths, areas, and volumes both for regular and irregular shapes from drawings. Theoretical processes of building contract from inception to completion and the interrelationship of the professional team Engineering drawings for buildings and various structures, Principles of

Aspect	What to prepare
	standard methods of measurement of work. Bills of quantities, Drawings. Purpose of Bills of Quantities. Processes of preparing Bills of Quantities, including taking off, working up, abstracting and billing. Types of bill formats and their uses. Setting out of descriptive and quantitative information in taking off. Dimension, abstracting and Billing Sheets means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Develop an idea on measurement of work 5 Understanding Historical development of quantity surveying: Functions performed by the quantity surveyor in relation to construction works - Evolution of standard methods of measurement for construction Works. The use of document in practice. Basic concepts of quantity surveying: Measurement and computation of lengths, girths, areas, and volumes both for regular and irregular shapes from drawings. Theoretical processes of building contract from inception to completion and the interrelationship of the professional team Engineering drawings for buildings and various structures, Principles of standard methods of measurement of work. Bills of quantities, Drawings. Purpose of Bills of Quantities. Processes of preparing Bills of Quantities, including taking off, working up, abstracting and billing. Types of bill formats and their uses. Setting out of descriptive and quantitative information in taking off. Dimension, abstracting and Billing Sheets □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

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description.
  M1.01        Explain basics of quantity surveying                    5
Remembering
  M1.02        Explain building estimates                              5
Remembering
  M1.03         Develop an idea on measurement of work                 5
Understanding

Contents:
Historical development of quantity surveying: Functions performed by the quantity
surveyor in relation to construction works - Evolution of standard methods of
measurement
for construction Works. The use of document in practice.
Basic concepts of quantity surveying: Measurement and computation of lengths,
girths,
areas, and volumes both for regular and irregular shapes from drawings.
Theoretical
processes of building contract from inception to completion and the
interrelationship of the
professional team Engineering drawings for buildings and various structures,
Principles of
standard methods of measurement of work. Bills of quantities, Drawings. Purpose
of Bills of
Quantities. Processes of preparing Bills of Quantities, including taking off,
working up,
abstracting and billing. Types of bill formats and their uses. Setting out of
descriptive and
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Comprehend analysis of Price and understand tendering and contracting

This module is presented from the official syllabus scope for Quantity Surveying. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Comprehend analysis of Price and understand tendering and contracting
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain price analysis of labours and construction

Explain price analysis of labours and construction is an official topic in Module 2 of Quantity Surveying. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain price analysis of labours and construction using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain price analysis of labours and construction should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain price analysis of labours and construction means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Explain price analysis of labours and construction □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Explain estimation and contracting and tendering 4 Applying Develop a solution for contract conditions and

Explain estimation and contracting and tendering 4 Applying Develop a solution for contract conditions and is an official topic in Module 2 of Quantity Surveying. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain estimation and contracting and tendering 4 Applying Develop a solution for contract conditions and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain estimation and contracting and tendering 4 applying develop a solution for contract conditions and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain estimation and contracting and tendering 4 Applying Develop a solution for contract conditions and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Explain estimation and contracting and tendering 4 Applying Develop a solution for contract conditions and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

5 Understanding administrations Analysis of prices of Construction materials, Labor, Machinery, Overheads & incidentals, Other incidentals in the analysis of rates for various trade items of civil engineering construction, Standard schedule of rates, Machinery and their uses in various construction activities, Output constants, Life, depreciation, cost of owning and operating charges for machinery, Economics of owning or hiring machinery. Estimating, tendering and contract management .Tender structure for different types of contracts, Tendering procedure, Specifications, Preparation of pre-qualification documents, Preparation of Tender Documents and invitation to bid, Submission of bids, Tender documents for LCB, NCB and ICB, Procurement of tender documents, Pre-bid conference and site investigation, Bank guarantees and Performance Bonds, Sub contracting &labour contracting system, Letter of Acceptance and Contract signing.

5 Understanding administrations Analysis of prices of Construction materials, Labor, Machinery, Overheads & incidentals, Other incidentals in the analysis of rates for various trade items of civil engineering construction, Standard schedule of rates, Machinery and their uses in various construction activities, Output constants, Life, depreciation, cost of owning and operating charges for machinery, Economics of owning or hiring machinery. Estimating, tendering and contract management .Tender structure for different types of contracts, Tendering procedure, Specifications, Preparation of pre-qualification documents, Preparation of Tender Documents and invitation to bid, Submission of bids, Tender documents for LCB, NCB and ICB, Procurement of tender documents, Pre-bid conference and site investigation, Bank guarantees and Performance Bonds, Sub contracting &labour contracting system, Letter of Acceptance and Contract signing. is an official topic in Module 2 of Quantity Surveying. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding administrations Analysis of prices of Construction materials, Labor, Machinery, Overheads & incidentals, Other incidentals in the analysis of rates for various trade items of civil engineering construction, Standard schedule of rates, Machinery and their uses in various

construction activities, Output constants, Life, depreciation, cost of owning and operating charges for machinery, Economics of owning or hiring machinery. Estimating, tendering and contract management .Tender structure for different types of contracts, Tendering procedure, Specifications, Preparation of pre-qualification documents, Preparation of Tender Documents and invitation to bid, Submission of bids, Tender documents for LCB, NCB and ICB, Procurement of tender documents, Pre-bid conference and site investigation, Bank guarantees and Performance Bonds, Sub contracting &labour contracting system, Letter of Acceptance and Contract signing. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding administrations analysis of prices of construction materials, labor, machinery, overheads & incidentals, other incidentals in the analysis of rates for various trade items of civil engineering construction, standard schedule of rates, machinery and their uses in various construction activities, output constants, life, depreciation, cost of owning and operating charges for machinery, economics of owning or hiring machinery. estimating, tendering and contract management .tender structure for different types of contracts, tendering procedure, specifications, preparation of pre-qualification documents, preparation of tender documents and invitation to bid, submission of bids, tender documents for lcb, ncb and icb, procurement of tender documents, pre-bid conference and site investigation, bank guarantees and performance bonds, sub contracting &labour contracting system, letter of acceptance and contract signing. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding administrations Analysis of prices of Construction materials, Labor, Machinery, Overheads & incidentals, Other incidentals in the analysis of rates for various trade items of civil engineering construction, Standard schedule of rates, Machinery and their uses in various construction activities, Output constants, Life, depreciation, cost of owning and operating charges for machinery, Economics of owning or hiring machinery. Estimating, tendering and contract management .Tender structure for different types of contracts, Tendering procedure, Specifications, Preparation of pre-qualification documents, Preparation of Tender Documents and invitation to bid, Submission of bids, Tender documents for LCB, NCB and ICB, Procurement of tender documents, Pre-bid conference and site investigation, Bank guarantees and Performance Bonds, Sub contracting &labour contracting system, Letter of Acceptance and Contract signing. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 5 Understanding administrations Analysis of prices of Construction materials, Labor, Machinery, Overheads & incidentals, Other incidentals in the analysis of rates for various trade items of civil engineering construction, Standard schedule of rates, Machinery and their uses in various construction activities, Output constants, Life, depreciation, cost of owning and operating charges for machinery, Economics of owning or hiring machinery. Estimating, tendering and contract management .Tender structure for different types of contracts, Tendering procedure, Specifications, Preparation of pre-qualification documents, Preparation of Tender Documents and invitation to bid, Submission of bids, Tender documents for LCB, NCB and ICB, Procurement of tender documents, Pre-bid conference and site investigation, Bank guarantees and Performance Bonds, Sub contracting &labour contracting system, Letter of Acceptance and Contract signing. □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Comprehend analysis of Price and understand tendering and contracting

  M2.01     Explain price analysis of labours and construction         5
Understanding

  M2.02     Explain estimation and contracting and tendering           4
Applying
            Develop a solution for contract conditions and
  M2.03                                                                5
Understanding
            administrations
            Series Test – I                                            1
Contents:
Analysis of prices of Construction materials, Labor, Machinery, Overheads &
incidentals,
Other incidentals in the analysis of rates for various trade items of civil
engineering
construction, Standard schedule of rates,
Machinery and their uses in various construction activities, Output constants,
Life,
depreciation, cost of owning and operating charges for machinery, Economics of
owning or
hiring machinery.
Estimating, tendering and contract management .Tender structure for different
types of
contracts, Tendering procedure, Specifications, Preparation of pre-qualification
documents,
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Summarize the land evaluation and surveyor laws

This module is presented from the official syllabus scope for Quantity Surveying. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize the land evaluation and surveyor laws
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain law of surveyors

Explain law of surveyors is an official topic in Module 3 of Quantity Surveying. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain law of surveyors using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain law of surveyors should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain law of surveyors means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Explain law of surveyors [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the assessment of property

Explain the assessment of property is an official topic in Module 3 of Quantity Surveying. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the assessment of property using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the assessment of property should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the assessment of property means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Explain the assessment of property [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain valuation of land and building 5 Applying Valuation - Explanation of terms, types of values, sinking fund, years purchase, Depreciation - Straight line method, constant percentage method, S.F method .Obsolescence. Valuation of real properties-rental method, profit based method, depreciation method. Valuation of landed properties -belting method, development method, hypothecated building scheme method. Rent calculation. Lease and Lease hold property Law of Surveyors- Valuation of land, buildings and machinery. The principal types of landed property, the incidents of their tenure, Income derived from them and the outgoing to which they are subjected; Laws governing valuation of land & building; Methods of valuation of Construction machinery.

Explain valuation of land and building 5 Applying Valuation - Explanation of terms, types of values, sinking fund, years purchase, Depreciation - Straight line method, constant percentage method, S.F method .Obsolescence. Valuation of real properties-rental method, profit based method, depreciation method. Valuation of landed properties -belting method, development method, hypothecated building scheme method. Rent calculation. Lease and Lease hold property Law of Surveyors- Valuation of land, buildings and machinery. The principal types of landed property, the incidents of their tenure, Income derived from them and the outgoing to which they are subjected; Laws governing valuation of land & building; Methods of valuation of Construction machinery. is an official topic in Module 3 of Quantity Surveying. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain valuation of land and building 5 Applying Valuation - Explanation of terms, types of values, sinking fund, years purchase, Depreciation - Straight line method, constant percentage method, S.F method .Obsolescence. Valuation of real properties-rental method, profit based method, depreciation method. Valuation of landed properties -belting method, development method, hypothecated building scheme method. Rent calculation. Lease and Lease hold property Law of Surveyors- Valuation of land, buildings and machinery. The principal types of landed property, the incidents of their tenure, Income derived

from them and the outgoing to which they are subjected; Laws governing valuation of land & building; Methods of valuation of Construction machinery. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain valuation of land and building 5 applying valuation - explanation of terms, types of values, sinking fund, years purchase, depreciation - straight line method, constant percentage method, s.f method .obsolescence. valuation of real properties-rental method, profit based method, depreciation method. valuation of landed properties -belting method, development method, hypothecated building scheme method. rent calculation. lease and lease hold property law of surveyors- valuation of land, buildings and machinery. the principal types of landed property, the incidents of their tenure, income derived from them and the outgoing to which they are subjected; laws governing valuation of land & building; methods of valuation of construction machinery. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain valuation of land and building 5 Applying Valuation - Explanation of terms, types of values, sinking fund, years purchase, Depreciation - Straight line method, constant percentage method, S.F method .Obsolescence. Valuation of real properties-rental method, profit based method, depreciation method. Valuation of landed properties -belting method, development method, hypothecated building scheme method. Rent calculation. Lease and Lease hold property Law of Surveyors- Valuation of land, buildings and machinery. The principal types of landed property, the incidents of their tenure, Income derived from them and the outgoing to which they are subjected; Laws governing valuation of land & building; Methods of valuation of Construction machinery. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain valuation of land and building 5 Applying Valuation - Explanation of terms, types of values, sinking fund, years purchase, Depreciation - Straight line method, constant percentage method, S.F method .Obsolescence. Valuation of real properties-rental method, profit based method, depreciation method. Valuation of landed properties -belting method, development method, hypothecated building scheme method. Rent calculation. Lease and Lease hold property Law of Surveyors- Valuation of land, buildings and machinery. The principal types of landed property, the incidents of their tenure, Income derived from them and the outgoing to which they are subjected; Laws governing valuation of land & building; Methods of valuation of Construction machinery. □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarize the land evaluation and surveyor laws

 M3.01      Explain law of surveyors                                  5
Remembering

 M3.02      Explain the assessment of property                        5
Applying

 M3.03      Explain valuation of land and building                    5
Applying

Contents:
Valuation - Explanation of terms, types of values, sinking fund, years purchase,
Depreciation - Straight line method, constant percentage method, S.F method
.Obsolescence.
Valuation of real properties-rental method, profit based method, depreciation
method.
Valuation of landed properties -belting method, development method, hypothecated
building scheme method. Rent calculation. Lease and Lease hold property Law of
Surveyors- Valuation of land, buildings and machinery.
The principal types of landed property, the incidents of their tenure, Income
derived from
them and the outgoing to which they are subjected; Laws governing valuation of land
&
building; Methods of valuation of Construction machinery.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Describe Project management methods and applications

This module is presented from the official syllabus scope for Quantity Surveying. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe Project management methods and applications
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

Explain project management

Explain project management is an official topic in Module 4 of Quantity Surveying. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain project management using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain project management should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain project management means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Explain project management [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Develop solutions for Project work breakdown 7 Applying Project Management - functions, processes, organization. Project manager’s role, responsibility, competency criteria; main causes of project failure. Project development process. Project management organization, Project planning and control approach. Types of scheduling : Barchart, Network, Line of Balance schedule, Time limited and Resources limited schedule, Schedule hierarchy, Resources forecasting, Manpower planning, Material planning, Material Procurement schedule, Planning and selection of equipment. Cost planning and Scheduling Computer applications in quantity surveying Quantity estimation, Tendering, Billing and Project Management concept; Quick and accurate quantity calculation; Text /Reference: T/R Book Title/Author Martin Brook, Estimating and tendering for construction work, T1 Butterworth-4th edition R2 Trever Holroyd, Principles of Estimating, Thomas Tesford, 2nd edition R3 Aliyanashworth, Practice and Procedure for Quantity Surveyors,Wileyindia R4 D DKohli, R CKohli.A Text book of Estimating and costing Online resources Sl. No Website Link 1 https://youtu.be/hfi59uBYacg https://www.quora.com/What-is-the-importance-and-purpose-of-quantity- 2 surveying-and-costing-in-construction-work-1 3 https://engineeringmanagementinstitute.org/quantity-surveyors-why-important/ 4 https://www.slideshare.net/Wallsasia/quantity-surveying-and-its-importance

Develop solutions for Project work breakdown 7 Applying Project Management - functions, processes, organization. Project manager’s role, responsibility, competency criteria; main causes of project failure. Project development process. Project management organization, Project planning and control approach. Types of scheduling : Barchart, Network, Line of Balance schedule, Time limited and Resources limited schedule, Schedule hierarchy, Resources forecasting, Manpower planning, Material planning, Material Procurement schedule, Planning and selection of equipment. Cost planning and Scheduling Computer applications in quantity surveying Quantity estimation, Tendering, Billing and Project Management concept; Quick and accurate quantity calculation; Text /Reference: T/R Book Title/Author Martin Brook, Estimating and tendering for construction work, T1 Butterworth-4th edition R2 Trever Holroyd, Principles of Estimating, Thomas Tesford, 2nd edition R3 Aliyanashworth, Practice and Procedure for Quantity Surveyors,Wileyindia R4 D DKohli, R CKohli.A Text book of Estimating and costing Online resources Sl. No Website Link 1 https://youtu.be/hfi59uBYacg https://www.quora.com/What-is-the-importance-and-purpose-of-quantity- 2 surveying-and-costing-in-construction-work-1 3 https://engineeringmanagementinstitute.org/quantity-surveyors-why-important/ 4 https://www.slideshare.net/Wallsasia/quantity-surveying-and-its-importance is an official topic in Module 4 of Quantity Surveying. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop solutions for Project work breakdown 7 Applying Project Management - functions, processes, organization. Project manager's role, responsibility, competency criteria; main causes of project failure. Project development process. Project management organization, Project planning and control approach. Types of scheduling : Barchart, Network, Line of Balance schedule, Time limited and Resources limited schedule, Schedule hierarchy, Resources forecasting, Manpower planning, Material planning, Material Procurement schedule, Planning and selection of equipment. Cost planning and Scheduling Computer applications in quantity surveying Quantity estimation, Tendering, Billing and Project Management concept; Quick and accurate quantity calculation; Text /Reference: T/R Book Title/Author Martin Brook, Estimating and tendering for construction work, T1 Butterworth-4th edition R2 Trever Holroyd, Principles of Estimating, Thomas Tesford, 2nd edition R3 Aliyanashworth, Practice and Procedure for Quantity Surveyors,Wileyindia R4 D DKohli, R CKohli.A Text book of Estimating and costing Online resources Sl. No Website Link 1 https://youtu.be/hfi59uBYacg https://www.quora.com/What-is-the-importance-and-purpose-of-quantity- 2 surveying-and-costing-in-construction-work-1 3 https://engineeringmanagementinstitute.org/quantity-surveyors-why-important/ 4 https://www.slideshare.net/Wallsasia/quantity-surveying-and-its-importance using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop solutions for project work breakdown 7 applying project management - functions, processes, organization. project manager's role, responsibility, competency criteria; main causes of project failure. project development process. project management organization, project planning and control approach. types of scheduling : barchart, network, line of balance schedule, time limited and resources limited schedule, schedule hierarchy, resources forecasting, manpower planning, material planning, material procurement schedule, planning and selection of equipment. cost planning and scheduling computer applications in quantity surveying quantity estimation, tendering, billing and project management concept; quick and accurate quantity calculation; text /reference: t/r book title/author martin brook, estimating and tendering for construction work, t1 butterworth-4th edition r2 trever holroyd, principles of estimating, thomas tesford, 2nd edition r3 aliyanashworth, practice and procedure for quantity surveyors,wileyindia r4 d dkohli, r ckohli.a text book of estimating and costing online resources sl. no website link 1 https://youtu.be/hfi59ubyacg https://www.quora.com/what-is-the-importance-and-purpose-of-quantity- 2 surveying-and-costing-in-construction-work-1 3 https://engineeringmanagementinstitute.org/quantity-surveyors-why-important/ 4 https://www.slideshare.net/wallsasia/quantity-surveying-and-its-importance should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop solutions for Project work breakdown 7 Applying Project Management - functions, processes, organization. Project manager’s role, responsibility, competency criteria; main causes of project failure. Project development process. Project management organization, Project planning and control approach. Types of scheduling : Barchart, Network, Line of Balance schedule, Time limited and Resources limited schedule, Schedule hierarchy, Resources forecasting, Manpower planning, Material planning, Material Procurement schedule, Planning and selection of equipment. Cost planning and Scheduling Computer applications in quantity surveying Quantity estimation, Tendering, Billing and Project Management concept; Quick and accurate quantity calculation; Text /Reference: T/R Book Title/Author Martin Brook, Estimating and tendering for construction work, T1 Butterworth-4th edition R2 Trever Holroyd, Principles of Estimating, Thomas Tesford, 2nd edition R3 Aliyanashworth, Practice and Procedure for Quantity Surveyors,Wileyindia R4 D DKohli, R CKohli.A Text book of Estimating and costing Online resources Sl. No Website Link 1 https://youtu.be/hfi59uBYacg https://www.quora.com/What-is-the-importance-and-purpose-of-quantity- 2 surveying-and-costing-in-construction-work-1 3 https://engineeringmanagementinstitute.org/quantity-surveyors-why-important/ 4 https://www.slideshare.net/Wallsasia/quantity-surveying-and-its-importance means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Develop solutions for Project work breakdown 7 Applying Project Management - functions, processes, organization. Project manager’s role, responsibility, competency criteria; main causes of project failure. Project development process. Project management organization, Project planning and control approach. Types of scheduling : Barchart, Network, Line of Balance schedule, Time limited and Resources limited schedule, Schedule hierarchy, Resources forecasting, Manpower planning, Material planning, Material Procurement schedule, Planning and selection of equipment. Cost planning and Scheduling Computer applications in quantity surveying Quantity estimation, Tendering, Billing and Project Management concept; Quick and accurate quantity calculation; Text /Reference: T/R Book Title/Author Martin Brook, Estimating and tendering for construction work, T1 Butterworth-4th edition R2 Trever Holroyd, Principles of Estimating, Thomas Tesford, 2nd edition R3 Aliyanashworth, Practice and Procedure for Quantity Surveyors,Wileyindia R4 D DKohli, R CKohli.A Text book of Estimating and costing Online resources Sl. No Website Link 1 https://youtu.be/hfi59uBYacg https://www.quora.com/What-is-the-importance-and-purpose-of-quantity- 2 surveying-and-costing-in-construction-work-1 3 https://engineeringmanagementinstitute.org/quantity-surveyors-why-important/ 4 https://www.slideshare.net/Wallsasia/quantity-surveying-and-its-importance □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Describe Project management methods and applications

 M4.01      Explain project management                                7
Remembering

 M4.02      Develop solutions for Project work breakdown              7
Applying

            Series Test – II                                          1

Contents:
Project Management - functions, processes, organization. Project manager’s role,
responsibility, competency criteria; main causes of project failure. Project
development
process. Project management organization, Project planning and control approach.
Types of scheduling : Barchart, Network, Line of Balance schedule, Time limited
and
Resources limited schedule, Schedule hierarchy, Resources forecasting, Manpower
planning,
Material planning, Material Procurement schedule, Planning and selection of
equipment.
Cost planning and Scheduling Computer applications in quantity surveying Quantity
estimation, Tendering, Billing and Project Management concept; Quick and accurate
quantity calculation;

Text /Reference:
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

- , Output constants, Life,
- depreciation, cost of owning and operating charges for machinery, Economics of owning or
- hiring machinery.
- Estimating, tendering and contract management .Tender structure for different types of
- contracts, Tendering procedure, Specifications, Preparation of pre-qualification documents, Preparation of Tender
- Documents and invitation to bid, Submission of bids, Tender documents for LCB, NCB and
- ICB, Procurement of tender documents, Pre-bid conference and site investigation, Bank
- guarantees and Performance Bonds, Sub contracting &labour contracting system, Letter of
- Acceptance and Contract signing.
- CO3 Summarize the land evaluation and surveyor laws
- M3.01 Explain law of surveyors 5 Remembering
- M3.02 Explain the assessment of property 5 Applying
- M3.03 Explain valuation of land and building 5 Applying
- Contents:
- Valuation - Explanation of terms, types of values, sinking fund, years purchase,
- Depreciation - Straight line method, constant percentage method, S.F method
- Obsolescence.
- Valuation of real properties-rental method, profit based method, depreciation method.
- Valuation of landed properties -belting method, development method, hypothecated
- building scheme method. Rent calculation. Lease and Lease hold property Law of

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
of property                        5      Applying

 M3.03      Explain valuation of land and building                  5      Applying

Contents:
Valuation - Explanation of terms, types of values, sinking fund, years purchase,
```

```
Depreciation - Straight line method, constant percentage method, S.F method
.Obsolescence.
Valuation of real properties-rental method, profit based method, depreciation method.
Valuation of landed properties -belting method, development method, hypothecated
building scheme method. Rent calculation. Lease and Lease hold property Law of
Surveyors- Valuation of land, buildings and machinery.
The principal types of landed property, the incidents of their tenure, Income derived
from
them and the outgoing to which they are subjected; Laws governing valuation of land &
building; Methods of valuation of Construction machinery.

  CO4       Describe Project management methods and applications

 M4.01      Explain project management                                7
Remembering

 M4.02      Develop solutions for Project work breakdown              7
Applying

            Series Test – II                                          1
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Explain basics of quantity surveying. (3 marks)

Answer: Begin with the standard definition or identity of *Explain basics of quantity surveying*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Explain building estimates. (3 marks)

Answer: Begin with the standard definition or identity of *Explain building estimates*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Develop an idea on measurement of work 5 Understanding Historical development of quantity surveying: Functions performed by the quantity surveyor in relation to construction works - Evolution of standard methods of measurement for construction Works. The use of document in practice. Basic concepts of quantity surveying: Measurement and computation of lengths, girths, areas, and volumes both for regular and irregular shapes from drawings. Theoretical processes of building contract from inception to completion and the interrelationship of the professional team Engineering drawings for buildings and various structures, Principles of standard methods of measurement of work. Bills of quantities, Drawings. Purpose of Bills of Quantities. Processes of preparing Bills of Quantities, including taking off, working up, abstracting and billing. Types of bill formats and their uses. Setting out of descriptive and quantitative information in taking off. Dimension, abstracting and Billing Sheets. (3 marks)

Answer: Begin with the standard definition or identity of *Develop an idea on measurement of work 5 Understanding Historical development of quantity surveying: Functions performed by the quantity surveyor in relation to construction works - Evolution of standard methods of measurement for construction Works. The use of document in practice. Basic concepts of quantity surveying: Measurement and computation of lengths, girths, areas, and volumes both for regular and irregular shapes from drawings. Theoretical processes of building contract from inception to completion and the interrelationship of the professional team Engineering drawings for buildings and various structures, Principles of standard methods of measurement of work. Bills of quantities, Drawings. Purpose of Bills of Quantities. Processes of preparing Bills of Quantities, including taking off, working up, abstracting and billing. Types of bill formats and their uses. Setting out of descriptive and quantitative information in taking off. Dimension, abstracting and Billing Sheets*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Explain price analysis of labours and construction. (3 marks)

Answer: Begin with the standard definition or identity of *Explain price analysis of labours and construction*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain estimation and contracting and tendering 4 Applying Develop a solution for contract conditions and. (3 marks)

Answer: Begin with the standard definition or identity of *Explain estimation and contracting and tendering 4 Applying Develop a solution for contract conditions and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Understanding administrations Analysis of prices of Construction materials, Labor, Machinery, Overheads & incidentals, Other incidentals in the analysis of rates for various trade items of civil engineering construction, Standard schedule of rates, Machinery and their uses in various construction activities, Output constants, Life, depreciation, cost of owning and operating charges for machinery, Economics of owning or hiring machinery. Estimating, tendering and contract management .Tender structure for different types of contracts, Tendering procedure, Specifications, Preparation of pre-qualification documents, Preparation of Tender Documents and invitation to bid, Submission of bids, Tender documents for LCB, NCB and

ICB, Procurement of tender documents, Pre-bid conference and site investigation, Bank guarantees and Performance Bonds, Sub contracting &labour contracting system, Letter of Acceptance and Contract signing.. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding administrations Analysis of prices of Construction materials, Labor, Machinery, Overheads & incidentals, Other incidentals in the analysis of rates for various trade items of civil engineering construction, Standard schedule of rates, Machinery and their uses in various construction activities, Output constants, Life, depreciation, cost of owning and operating charges for machinery, Economics of owning or hiring machinery. Estimating, tendering and contract management .Tender structure for different types of contracts, Tendering procedure, Specifications, Preparation of pre-qualification documents, Preparation of Tender Documents and invitation to bid, Submission of bids, Tender documents for LCB, NCB and ICB, Procurement of tender documents, Pre-bid conference and site investigation, Bank guarantees and Performance Bonds, Sub contracting &labour contracting system, Letter of Acceptance and Contract signing.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Explain law of surveyors. (3 marks)

Answer: Begin with the standard definition or identity of *Explain law of surveyors*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the assessment of property. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the assessment of property*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain valuation of land and building 5 Applying Valuation - Explanation of terms, types of values, sinking fund, years purchase, Depreciation - Straight line method, constant percentage method, S.F method .Obsolescence. Valuation of real properties-rental method, profit based method, depreciation method. Valuation of landed properties -belting method, development method, hypothecated building scheme method. Rent calculation. Lease and Lease hold property Law of Surveyors- Valuation of land, buildings and machinery. The principal types of landed property, the incidents of their tenure, Income derived from them and the outgoing to which they are subjected; Laws governing valuation of land & building; Methods of valuation of Construction machinery.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain valuation of land and building 5 Applying Valuation - Explanation of terms, types of values, sinking fund, years purchase, Depreciation - Straight line method, constant percentage method, S.F method .Obsolescence. Valuation of real properties-rental method, profit based method, depreciation method. Valuation of landed properties -belting method, development method, hypothecated building scheme method. Rent calculation. Lease and Lease hold property Law of Surveyors- Valuation of land, buildings and machinery. The principal types of landed property, the incidents of their tenure, Income derived from them and the outgoing to which they are subjected; Laws governing valuation of land & building; Methods of valuation of Construction machinery.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Explain project management. (3 marks)

Answer: Begin with the standard definition or identity of *Explain project management*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Develop solutions for Project work breakdown 7 Applying Project Management - functions, processes, organization. Project manager’s role, responsibility, competency criteria; main causes of project failure. Project development process. Project management organization, Project planning and control approach. Types of scheduling : Barchart, Network, Line of Balance schedule, Time limited and Resources limited schedule, Schedule hierarchy, Resources forecasting, Manpower planning, Material planning, Material Procurement schedule, Planning and selection of equipment. Cost planning and Scheduling Computer applications in quantity surveying Quantity estimation, Tendering, Billing and Project Management concept; Quick and accurate quantity calculation; Text /Reference: T/R Book Title/Author Martin Brook, Estimating and tendering for construction work, T1 Butterworth-4th edition R2 Trever Holroyd, Principles of Estimating, Thomas Tesford, 2nd edition R3 Aliyanashworth, Practice and Procedure for Quantity Surveyors,Wileyindia R4 D DKohli, R CKohli.A Text book of Estimating and costing Online resources Sl. No Website Link 1 https://youtu.be/hfi59uBYacg https://www.quora.com/What-is-the-importance-and-purpose-of-quantity- 2 surveying-and-costing-in-construction-work-1 3 https://engineeringmanagementinstitute.org/quantity-surveyors-why-important/ 4 https://www.slideshare.net/Wallsasia/quantity-surveying-and-its-importance. (3 marks)

Answer: Begin with the standard definition or identity of *Develop solutions for Project work breakdown 7 Applying Project Management - functions, processes, organization. Project manager’s role, responsibility, competency criteria; main causes of project failure. Project development process. Project management organization, Project planning and control approach. Types of scheduling : Barchart, Network, Line of Balance schedule, Time limited and Resources limited schedule, Schedule hierarchy, Resources forecasting, Manpower planning, Material planning, Material Procurement schedule, Planning and selection of equipment. Cost planning and Scheduling Computer* *applications in quantity surveying Quantity estimation, Tendering, Billing and Project Management concept; Quick and accurate quantity calculation; Text /Reference: T/R Book Title/Author Martin Brook, Estimating and tendering for construction work, T1 Butterworth-4th edition R2 Trever Holroyd, Principles of Estimating, Thomas Tesford, 2nd edition R3 Aliyanashworth, Practice and Procedure for Quantity Surveyors,Wileyindia R4 D DKohli, R CKohli.A Text book of Estimating and costing Online resources Sl. No Website Link 1 https://youtu.be/hfi59uBYacg https://www.quora.com/What-is-the-importance-and-purpose-of-quantity- 2 surveying-and-costing-in-construction-work-1 3 https://engineeringmanagementinstitute.org/quantity-surveyors-why-important/ 4 https://www.slideshare.net/Wallsasia/quantity-surveying-and-its-importance*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain basics of quantity surveying**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Explain price analysis of labours and construction**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain law of surveyors**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain project management**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No Website Link https://youtu.be/hfi59uBYacg
- https://www.quora.com/What-is-the-importance-and-purpose-of-quantity- surveying-and-costing-in-construction-work-1
- https://engineeringmanagementinstitute.org/quantity-surveyors-why-important/
- https://www.slideshare.net/Wallsasia/quantity-surveying-and-its-importance

IN-PAGE SEARCH

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